usb_power

File: usb_power.kicad_sch

USB-C sink (no host), VBUS protection, USB0 device/FEL

power

File: power.kicad_sch

Regulators, rail order, supervisors, staged-power links

f133

File: f133.kicad_sch

F133-B SoC: supplies, clock, reset, pin assignment

boot_nor

File: boot_nor.kicad_sch

32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery

mn820

File: mn820.kicad_sch

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel

fpga

File: fpga_ecp5.kicad_sch

LFE5U-45F-6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation

psram

File: psram.kicad_sch

APS6404L 8 MiB PSRAM: switched VDD (DONE-gated), static pulls

sd_audio

File: sd_audio.kicad_sch

microSD game media (not boot), 2x MAX98357A I2S speakers

controls

File: controls.kicad_sch

Game buttons, LEDs, debug UART

NOT approved for fabrication. LCD/panel/backlight deferred to R09.

Sheet: /
File: ARC1_R85.kicad_sch

Title: ARC-1 Compact R8.5 — hierarchical root

Size: A3	Date: 2026-10-15	**Rev: R85-draft**
KiCad E.D.A. 9.0.9		Id: 1/10

USB-C sink (no host), VBUS protection, USB0 device/FEL
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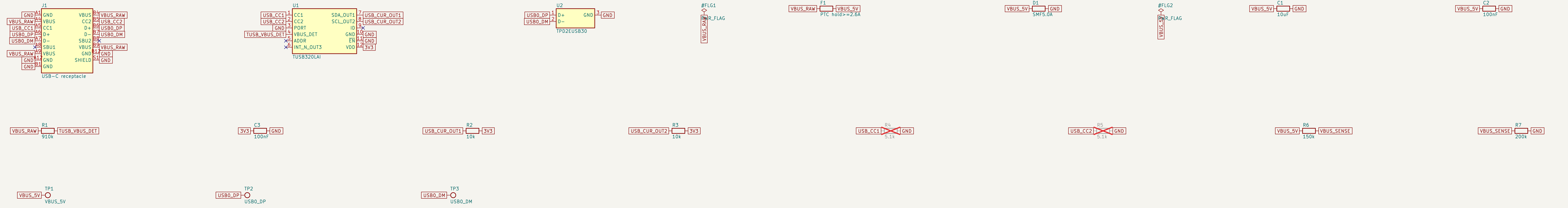


Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1

Sheet: /usb_power/
File: usb_power.kicad_sch

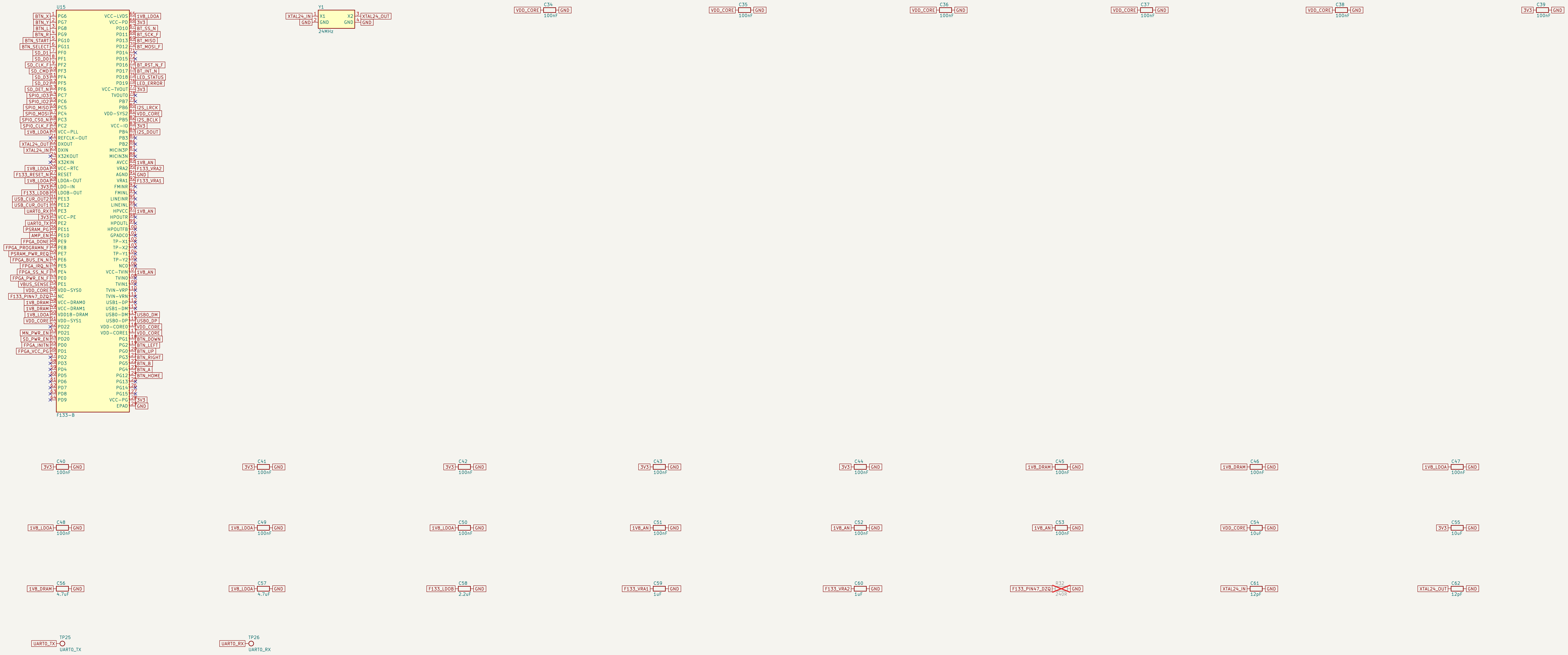
Title: ARC-1 Compact R8.5: USB-C sink (no host), VBUS protection, USB0 d

Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 2/10

Regulators, rail order, supervisors, staged−power links
Sheet generated from design.py; global labels carry every net.

Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC−1
Sheet: /power/
File: power.kicad_sch
Title: ARC−1 Compact R8.5: Regulators, rail order, supervisors, staged−powe
Size: User
Date: 2026−10−15
Rev: R85−draft
KiCad E.D.A. 9.0.9
Id: 3/10

F133-B SoC: supplies, clock, reset, pin assignment
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
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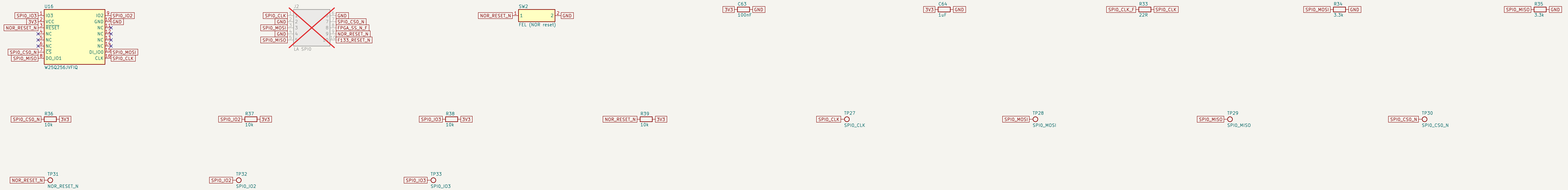
STRUTHIO ARC-1
Sheet: /f133/
File: f133.kicad_sch
Title: ARC-1 Compact R8.5: F133-B SoC: supplies, clock, reset, pin assignm
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 4/10

32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication

STRUTHIO ARC-1

Sheet: /boot_nor/
File: boot_nor.kicad_sch

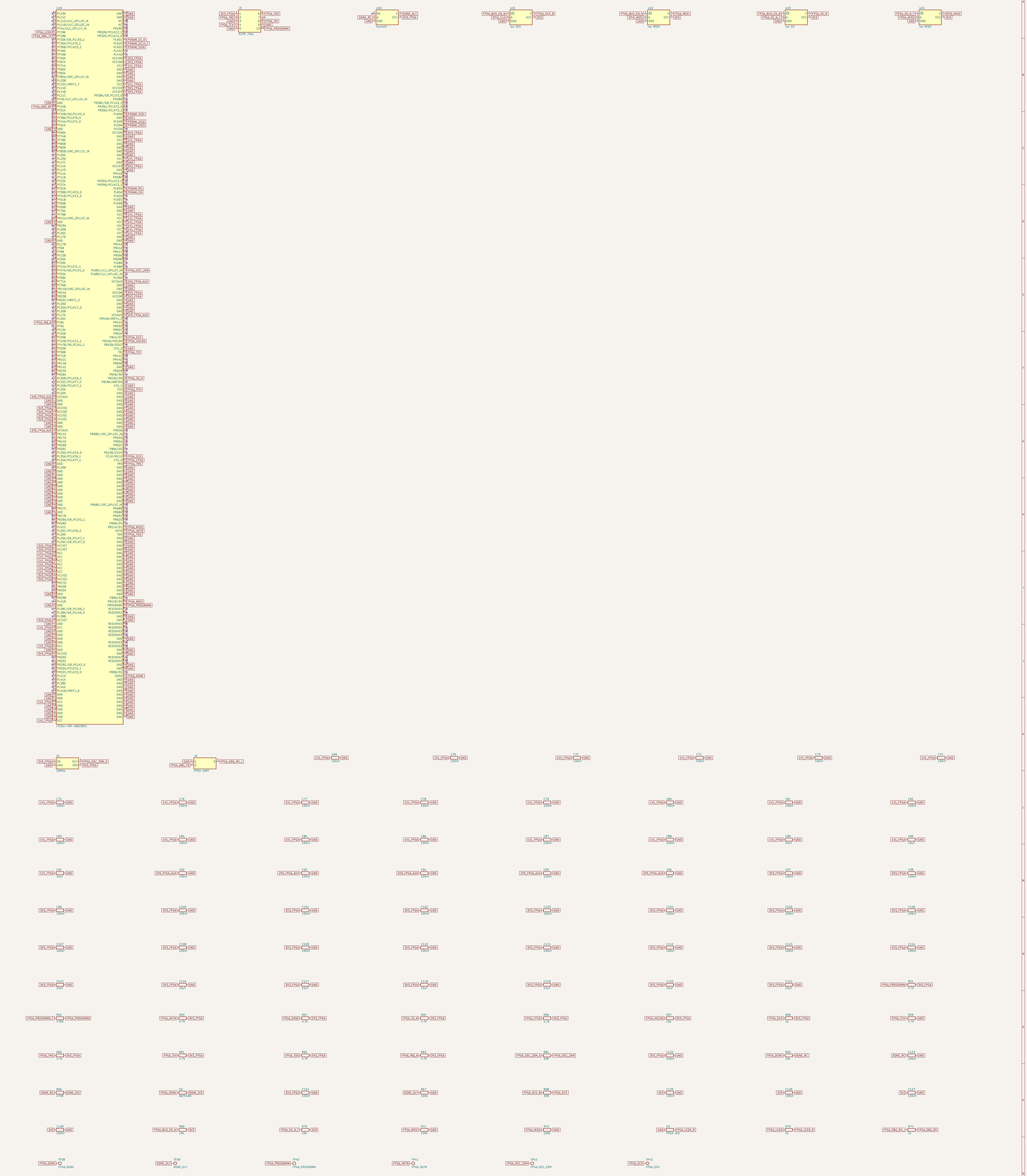
Title: ARC-1 Compact R8.5: 32 MiB SPI NOR boot (BOOT-SEL 00), straps, Fl

Size: User	Date: 2026-10-15	**Rev: R85-draft**
KiCad E.D.A. 9.0.9		Id: 5/10

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel
Sheet generated from design.py; global labels carry every net.

Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /mn820/
File: mn820.kicad_sch
Title: ARC-1 Compact R8.5: MN820 (BT820B + 128 MiB DDR3L) module sock
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 6/10

LFE5U-45F-6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1

Sheet: /fpga/
File: fpga_ecp5.kicad_sch

Title: ARC-1 Compact R8.5: LFE5U-45F-6BG381C (optional accelerator): rai

Size: User	Date: 2026-10-15	**Rev: R85-draft**
KiCad E.D.A. 9.0.9		Id: 7/10

APS6404L 8 MiB PSRAM: switched VDD (DONE–gated), static pulls
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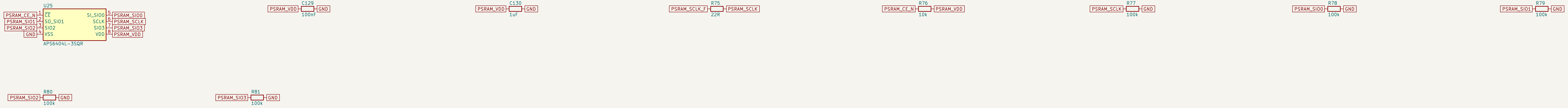
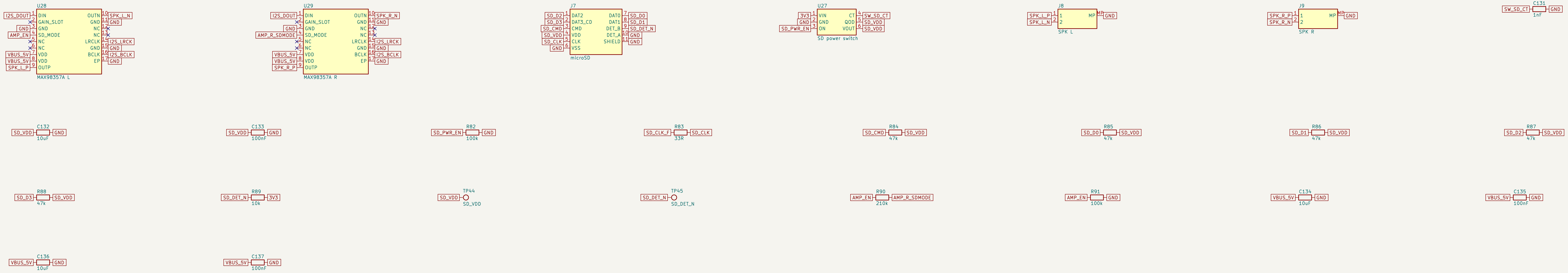


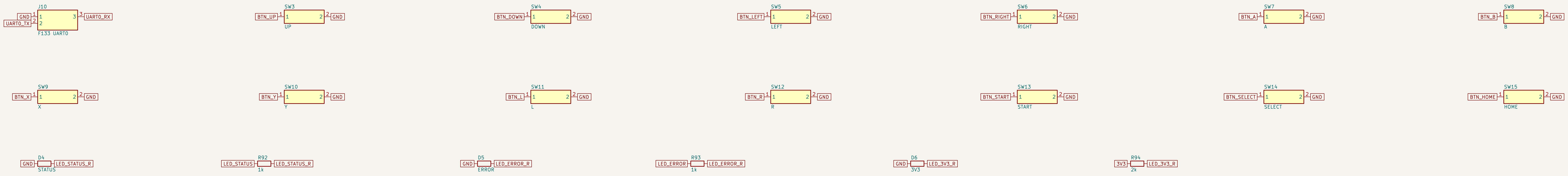
Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC–1
Sheet: /psram/
File: psram.kicad_sch
Title: ARC–1 Compact R8.5: APS6404L 8 MiB PSRAM: switched VDD (DONE–g
Size: User Date: 2026–10–15 **Rev: R85–draft**
KiCad E.D.A. 9.0.9 Id: 8/10

microSD game media (not boot), 2x MAX98357A I2S speakers
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /sd_audio/
File: sd_audio.kicad_sch
Title: ARC-1 Compact R8.5: microSD game media (not boot), 2x MAX98357A
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 9/10

Game buttons, LEDs, debug UART
Sheet generated from design.py: global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /controls/
File: controls.kicad_sch
Title: ARC-1 Compact R8.5: Game buttons, LEDs, debug UART
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 10/10